

YEAR 7 • UNIT 1 • HOMEWORK 4

Squares, Cubes and Roots

Mathematics 1.6 — Square Roots and Cube Roots • 1.5 — Tests for Divisibility

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

- Write your answers **on this sheet**. **No calculator**.
- Always **write the calculation, not just the answer**.
- Do **A1** first. You need those two lists for the whole packet.
- In **Section B** you use last week's divisibility tests to find roots.

Section A • Mathematics 1.6 — Squares and Cubes

Remember from the lesson

5^2 means $5 \times 5 = 25$. We say "**five squared**". The small 2 counts **how many fives**, so 5^2 is **not** 5×2 . 5^3 means $5 \times 5 \times 5 = 125$. We say "**five cubed**".

A **root** goes **backwards**. $5^2 = 25$, so the **square root** of 25 is 5. $5^3 = 125$, so the **cube root** of 125 is 5.

A1 The two lists

(a) Fill in the **twelve square numbers**. Use your notebook.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2												

(b) Now the **cube numbers**.

n	1	2	3	4	5	6	10
n^3							

(c) One number is on **both** lists. Which one? _____

A2 Square it, cube it, root it

(a) Work out **both**. Then write which one is bigger.

$7^2 = \underline{\hspace{2cm}}$

$7 \times 2 = \underline{\hspace{2cm}}$

Bigger: _____

(b) Use your lists from A1.

(i) $11^2 = \underline{\hspace{2cm}}$

(iv) $10^3 = \underline{\hspace{2cm}}$

(vii) $\sqrt[3]{27} = \underline{\hspace{2cm}}$

(ii) $12^2 = \underline{\hspace{2cm}}$

(v) $\sqrt{81} = \underline{\hspace{2cm}}$

(viii) $\sqrt[3]{216} = \underline{\hspace{2cm}}$

(iii) $4^3 = \underline{\hspace{2cm}}$

(vi) $\sqrt{144} = \underline{\hspace{2cm}}$

(ix) $\sqrt[3]{64} - \sqrt{100} = \underline{\hspace{2cm}}$

(c) One answer in (b) is **negative**. Which one? _____

A3 Say it, then write it**Word help – these words tell you what to do**

square 9

$9^2 = 9 \times 9$

the square of 9

the same thing: $9^2 = 81$

the square root of 81

9, because $9 \times 9 = 81$

cube 4

$4^3 = 4 \times 4 \times 4$

the cube root of 64

4, because $4 \times 4 \times 4 = 64$

a square number

a number on your first list

consecutive

one after another, nothing missed out. 6 and 7 are consecutive. 36 and 49 are consecutive **square** numbers.

Write the calculation for each sentence, then work it out. **Calculation first.**

1. Square 11.

Calculation: _____ Answer: _____

2. Find the square root of 121.

Calculation: _____ Answer: _____

3. Subtract the cube root of 27 from the square of 8.

Calculation: _____ Answer: _____

A4 Trap the root**When the root is not a whole number**

45 is not a square number, so $\sqrt{45}$ is not whole. But 45 is between **36** and **49**, and those two **are** on the list. So $\sqrt{45}$ is between **6** and **7** – two **consecutive** whole numbers.

Write the two consecutive whole numbers each root is between.

(a) _____ $< \sqrt{30} <$ _____

(c) _____ $< \sqrt{110} <$ _____

(b) _____ $< \sqrt{70} <$ _____

(d) _____ $< \sqrt[3]{50} <$ _____

(e) Two consecutive square numbers add up to 85. Which two? _____

Their square roots are _____ and _____

Section B · Maths 1.5 and 1.6 – Using the Tests on a Root**A square is a pair. A cube is three copies.**

$36 = 6 \times 6$. Both numbers in the pair are the **same**, so every factor inside one is inside the other as well. If **3** divides a square number, then **9** divides it too.

A cube is **three** copies: $8 = 2 \times 2 \times 2$. So a factor is in there **three times**. If **2** divides a cube number, then **8** divides it too.

Rule 1 divides by 3 but **not** by 9 $\Rightarrow$ **not** a square number

Rule 2 divides by 2 but **not** by 4 $\Rightarrow$ **not** a square number

Rule 3 divides by 2 but **not** by 8 $\Rightarrow$ **not** a cube number

Careful. These rules can only say **NO**. If a number passes, that does **not** prove it is a square.

B1 Can it be a square number?

Use the tests and fill in every box. In the last column write **NO** if a rule says no, or **maybe** if it passes. If the number is **odd**, put a dash in the **by 2** and **by 4** columns.

Number	Digit sum	by 3?	by 9?	by 2?	by 4?	Square?
132						
441						
162						
1296						
108						

(g) One number passed **every** rule. Which one? _____

Check it against your A1 list. It is between _____ and _____ – two square numbers.

So is it a square number? _____ (yes or no)

B2 Split it, then find the root**Split the number into two pieces you know**

Split the number into two pieces. Find the root of each piece. Then **multiply the two roots**. **Good pieces:** 9, 4 and 100 are squares (roots 3, 2, 10); 8 and 1000 are cubes (roots 2 and 10).

Example. $\sqrt{576}$. The digits add to $5 + 7 + 6 = 18$, so **9 divides it**. $576 \div 9 = 64$. Now $\sqrt{9} = 3$ and $\sqrt{64} = 8$, so $\sqrt{576} = 3 \times 8 = 24$. Check: $24 \times 24 = 576$.

Now do these. The number to divide by is given. Fill in the rest.

Find	Divide by	Split it	The two roots	Answer
$\sqrt{441}$	9			
$\sqrt{1296}$	9			
$\sqrt[3]{1728}$	8			
$\sqrt[3]{8000}$	1000			

B3 The challenge: 5832**5832, from Homework 3**

In Homework 3 you ticked **5832** for 2, 3, 4, 6, 8 and 9. So it passes **every** rule in B1 – but it is **not** a square number. It **is** a cube number. Find its cube root.

Step 1. 5832 divides by 8. Do the division. **Tip: dividing by 8 is the same as halving three times.** $5832 \div 8 =$ _____

Step 2. Your answer to Step 1 is $9 \times 9 \times 9$, so its cube root is _____

Step 3. $\sqrt[3]{8} =$ _____, so $\sqrt[3]{5832} =$ _____ $\times$ _____ $=$ _____

Step 4. Check your answer. $18 \times 18 =$ _____ then _____ $\times 18 =$ _____

B4 Word problem

The photograph. Mr Bowen puts **400 students** in a **perfect square**. The photographer climbs onto the roof. How many students are in each row? Which test did you use to split 400?

Answer Key – Teacher Copy

A1 The two lists

(a) 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

(b) 1, 8, 27, 64, 125, 216, and $10^3 = 1000$.

(c) $64 - 8 \times 8$ and $4 \times 4 \times 4$.

Mark this one first and mark it hard. A student who cannot fill these two rows cannot do Section B at all, because Section B is nothing but splitting a big number into two entries from these lists. If more than a few of the class leave gaps, re-copy the lists at the board before going through anything else in the packet.

A2 Out, and back again

(a) $7^2 = 49$, $7 \times 2 = 14$. 7^2 is bigger.

(b) (i) 121 (ii) 144 (iii) 64 (iv) 1000 (v) 9 (vi) 12 (vii) 3 (viii) 6 (ix) $4 - 10 = -6$

(c) (ix), which comes to -6 .

(a) is the whole reason this question exists. An answer of 14 for 7^2 is the small-2-means-times-2 error and it is worth ten seconds at the board, not a red cross. In (b), (ix) is the one they will call wrong: the class has decided a root question cannot produce a negative, and $4 - 10$ says otherwise — that is 1.1 arriving inside a 1.6 question. Part (c) exists to make them notice it rather than quietly change it to 6.

A3 Say it, then write it

1. $11^2 = 11 \times 11 = 121$

2. $\sqrt{121} = 11$

3. $8^2 - \sqrt[3]{27} = 64 - 3 = 61$

Question 3 is the English question, not the arithmetic one. *Subtract A from B* means $B - A$, so the calculation is $64 - 3$. A student who writes $3 - 64$ has read the sentence left to right and has made the same error they made with subtraction in 1.1 — send them back to re-read the sentence rather than marking the arithmetic. Questions 1 and 2 are on the page next to each other on purpose: same numbers, opposite directions.

A4 Trap the root

(a) $5 < \sqrt{30} < 6$ (b) $8 < \sqrt{70} < 9$ (c) $10 < \sqrt{110} < 11$ (d) $3 < \sqrt[3]{50} < 4$

(e) **36 and 49**. Their roots are **6** and **7**.

(d) is the one to check, because it is the only cube: they need $27 < 50 < 64$, and a student running on the square list will answer 7 and 8. (e) is the *consecutive* question. Consecutive square numbers are 11 apart here, not 1 apart, and a student who reads consecutive as “one apart” will hunt for two numbers differing by 1 and give up. If several fail it, that is a vocabulary lesson for Monday, not a marking problem.

B1 Can it be a square number?

132	sum 6	yes	no	yes	yes	NO
441	sum 9	yes	yes	—	—	maybe (it is 21^2)
162	sum 9	yes	yes	yes	no	NO
1296	sum 18	yes	yes	yes	yes	maybe (it is 36^2)
108	sum 9	yes	yes	yes	yes	maybe — but it is not

(g) **108**, between **100** and **121** (10^2 and 11^2). Not a square number.

The two rows that teach are **162** and **108**, and they teach opposite things. 162 passes the 3-and-9 rule and is killed by the 2-and-4 one, so a student who only ran the digit sum gets it wrong — both rules, every time. 108 is the honest one: it passes everything and is still not a square, which is the difference between a test and a proof. This used to be an open question asking students to explain that in their own words, which was too abstract for the class; it is now the same trap-the-root check as A4, applied to 108 instead of a root. If a student answers “yes” here, send them back to count: $10 \times 10 = 100$ and $11 \times 11 = 121$, and 108 is neither. A “maybe” written for 441 and 1296 in the table is the correct answer even though both really are squares — do not accept “yes” for those two on the strength of the rules alone.

B2 Split it, then find the root

$\sqrt{441}$	$\div 9$	9×49	3 and 7	21
$\sqrt{1296}$	$\div 9$	9×144	3 and 12	36
$\sqrt[3]{1728}$	$\div 8$	8×216	2 and 6	12
$\sqrt[3]{8000}$	$\div 1000$	1000×8	10 and 2	20

Divisions worth having to hand: $441 \div 9 = 49$, $1296 \div 9 = 144$, $1728 \div 8 = 216$, $8000 \div 1000 = 8$.

The divisor is now printed for every row, which used to be the hardest part — deciding whether to use 9, 8 or 1000 was one guess before the maths even started, and that made the question feel like four separate puzzles rather than one method practised four times. With the divisor given, this is the question the packet was built around: $\sqrt{1296}$ shows whether it has landed, since 1296 is nowhere near their A1 list and dividing by 9 is the only route to 36. The error to expect is **adding** the roots instead of multiplying them — $3 + 12 = 15$; tell them to check by squaring, because $15^2 = 225$ and not 1296.

B3 The challenge: 5832

Step 1: $5832 \div 8 = 729$ (halve three times: $5832 \rightarrow 2916 \rightarrow 1458 \rightarrow 729$).

Step 2: $\sqrt[3]{729} = 9$.

Step 3: $\sqrt[3]{8} = 2$, so $\sqrt[3]{5832} = 2 \times 9 = 18$.

Step 4: $18 \times 18 = 324$, and $324 \times 18 = 5832$.

Worth doing at the board even if most of the class got it, because it is the entire unit in one question: a divisibility test from 1.5 picks the split, a division carries it out, a cube root from 1.6 finishes it, and the check in Step 4 is two two-digit multiplications — which is exactly what the extra sheet is for. The halving tip in Step 1 matters: $5832 \div 8$ by long division is the hardest single step in the packet, and most students have not thought to use $\div 2, \div 2, \div 2$ instead. Step 4 is now two separate blanks rather than one open box, so a student who gets 324 but stalls on 324×18 has still shown the harder half of the check.

This question is also the answer to B1(g) arriving in a form nobody can argue with. 5832 passed every rule and is not a square, so if the class was unconvinced earlier, come back to B1(g) after this rather than re-teaching it cold.

B4 Word problem

$\sqrt{400} = 20$, so **20 students in each row**. The split: 400 ends in 00, so $400 = 4 \times 100$, and $\sqrt{4} = 2$, $\sqrt{100} = 10$, giving $2 \times 10 = 20$.

A student who simply knows that $20 \times 20 = 400$ has the right answer — but the question asked which test they used, so ask for the split as well. This is the only place in the packet where a square root has to be dug out of an English sentence, so mark the reading as carefully as the arithmetic: the phrase doing the work is *a perfect square*, and a student who multiplies 400×400 has not found it. The photographer on the roof is the deadpan line. Read it straight and move on.